

Active Inference LiveStream 054.2 ~ "...Compositional Account of the Bayesian Brain"

Livestream | (Smithe) | 1:41:58

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All right, we're in livestream 54.2, continuing with our third discussion on the work Mathematical Foundations for a Compositional Account of the Bayesian Brain. We're back with Dean, Ali, and Toby. So if each want to say hello and anything to begin with, otherwise Ali has some questions ready and we have some questions that people have submitted in the meantime as well. So, shall we just proceed to the questions as we have the same people as .1? Great. Okay, I'm going to jump into some of the pre-submitted questions. So, would it be correct to call equation 4.2, Compositional Bayes' Theorem, which is or are Bayes' Theorem categorically? And we have the document ready or anything that you want to share. But what does it mean for Bayes' Theorem to be categorified? And where is that in the dissertation? Yeah, okay. So that's a nice question. I would say that equation 4.2 is like a sort of basic representation of the product law of probability probability theory. And so I wouldn't say it's particularly like inherently compositional. It's just something that is just like an axiom that is just like a consequence of the sort of standard axioms of probability theory. The thing that makes it a little bit different from the way that it's normally written in probability theory is that I've kind of annotated those P symbols which represent like the probability of a certain event. I've annotated them each with like a subscript and that subscript represents like where the probability is coming from or like the probability according to whom. Well, that's kind of how you can think of it. And so normally I find like this kind of like P notation to be a little bit confusing because people often write, you know, P of B given A and they also write like P of like X given Y or like P of W comma A comma Z . And so like in each of those cases like the sort of content of the P is kind of different. Like the A and the B might be like events of a different kind to X and Y and yet they're all like put into this P symbol. Whereas in this case, like with the annotations it says like, as I say, it says according to what? And so here like the little subscript C represents what's typically called in like compositional probability theory like a channel or like a kernel. But that is like a kind of abstract way of just saying conditional probability distribution. And so the little c means it's a conditional probability distribution from like X , a space X to Y , a space Y . And here I'm thinking of like the B as being an event of type Y and A as being an event of type A , sorry, of type X . And so the PC says it's the probability of the event B of type Y given the event A of type X . And so that means that when you like plug these like these kind of flows of information together, you make sure like the types match. It's kind of like working with

a strongly typed programming language. So like that type information is like the main thing that's different between this and the standard like product rule of probability. And it's from that product rule of probability theory that you can just write down Bayes' theorem, which is the equation 4.3.

I would say that the reason why I slightly hesitate to call equation 4.3 like a sort of abstract or categorical version of Bayes' theorem, even though it's annotated by these channels, it's because you have this like division. And like that division isn't something that's like well defined in all categories, in all situations that you might want to consider like probability theory. For instance, like you might have this E of C dot π thing, and that means the kind of like Y marginal of the joint distribution, or it's otherwise known as the push forward of the distribution π through the channel C , like that thing might have what's known as measure zero, which means the probability might be zero. And so the division isn't very well defined. And so like that expression doesn't always make sense. Now, the thing that does always make sense, on the other hand, is that just product law equation, which tells you how the two ways you can write down this joint distribution. So if you have a joint distribution, like on the space X times Y , I've written it there, then you can decompose it into like a prior on X , and a channel from X to Y . So then if you like, put, you know, the information from the prior through the channel, you get the marginal on Y , and that's that push forward thing. Or you could write the joint distribution as a prior on Y , and a conditional distribution from Y to X . And the thing is, obviously, we're talking about the same joint distribution in both cases. So those two decompositions, they're sometimes called disintegrations, those two disintegrations need to be equal. And that's what that equation is saying. So that's kind of because it doesn't have this division in it. And because Bayes law follows from it, and because it's something you can write down in all settings, it's kind of more fundamental. But I quite like the kind of string diagrammatic form of it, which is something that is drawn, it's not it's not my invention, it's I think I originally saw it in a paper by Bart Jacobs, and I think his student at the time, Kenta Cho. So in, let's see, let's see, equation 4.5. So that's section 4.1.2 on abstract Bayesian inversion. So yeah, if you scroll down a little, it's on like page 130. Or 130 in my version, which might be hopefully around the same place in your version. Yeah, a little further.

Oh, man, no, maybe not. Let me let me let me get up the version on, because I've realized I've opened the wrong version.

Meanwhile, see if

So the thing I'm going for here is the sort of, as I say, the string diagrammatic depiction. Oh, yeah, 4.1.2. It's page 120. Sorry about that.

Oh, good.

Yeah, so it kind of shows you like how the information flows. And so this like makes the two, the two like processes, like quite explicit, like so here on the left, the prior of the joint distribution is π . And then you take the sort of probability,

the information from π , and you copy it, and you keep one of the copies, and that gives you your x marginal, and then you feed the other one through the likelihood or the conditional channel, c , and that gives you the other marginal. And so like, if you if you throw away this x information, you get the y marginal, and that's what marginalization is. And that's why the y marginal, if you look on the right hand side, the prior is c after π , you know, that's, that's the y marginal. So like, that's, that's, this is, this is like depicting that product law. And then you say, you know, you can call something a Bayesian inversion of c with respect to the prior π , if it is something that can go in the box on the right hand side where I've written $c \dashv \pi$. And the reason I use this dagger symbol is because under, you know, certain, you know, technical conditions, this, this Bayesian inversion thing gives you what's known in category theory as a dagger functor. And like, that's something that like swaps the direction of a morphism. And so here, you've taken the thing that goes from x to y , and you swapped it, and it goes from y to x . But it depends on the prior. And so I pretend that's all subscript. And so for me, this picture is a kind of, I mean, it's kind of equivalent to that product law equation 4.2. But I think it's, it's kind of, although it's a kind of new kind of language, I think it's a kind of more intuitive one, because you can sort of see how the information flows through the system. So that was a bit of a long answer. But I think that's like, I think that's how I think of it. Great. Okay. Next question could be a short one. I'm mostly wondering about how chapter six will be rewritten. Yeah, yeah. Will it be major? And when we promise we didn't write this question, primarily to select the best policy as to where to direct my regime of attention. Like, should I wait to dig deeper into chapter six until the updated observations arrive? Is reflecting deeply up until chapter five, where I should focus for now? Or is chapter six something that won't change too much? And I can still explore that next week. So this is a great question. And I would, and I'm sorry that it isn't like the new, the rewritten version isn't online yet. But it is really close, I promise. I am kind of aiming for it to be done, like next week, or maybe let's, let's not hold myself hostage to fortune. So let's call it the next couple of weeks. But it is a quite a, it's quite a change in the sense that I've kind of both, it's kind of annoying, because it's simpler, but also kind of more structured. And so that means that there's kind of, there were a lot of like finicky details that I didn't anticipate needing to deal with, that I had to deal with, even though fundamentally, the basic idea is kind of the same, but also neater and simpler at the same time. And if you, I mean, I can go into some details about the differences, if you'd like, or I can just leave it at that. Like, what would you rather? Let's return to this in the dot three.

Okay, yeah, let's do that. That's, that's a nice idea. So yeah, I would say don't, you know, dig too deeply into it. Because basically, the thing that I've managed to do away with, and sorry for all the people who tried to bang their heads on it already, is the kind of pro functor, like optics, kind of like the sort of pro functor stuff with those weird, like co end things. The kind of like integral symbols that I was using to define contexts for statistical games. That whole thing, you it turns out you don't actually need the complexity of that structure. And the notion of context is really given just by two things. It's really just the prior,

which we've already talked about. And also like the observation you take from the environment.

And so before I thought you needed to consider like not just the observation, but also like the stuff that might go on in the environment around the observation that like, gives you correlations. But it turns out actually, you don't need that, at least not for the free energy principle.

And so that makes things a bit simpler, because you don't need that kind of weird structure.

But there's like, as I say, like some technical stuff, which is very annoying, which made it also a little bit like slightly annoying to write down. And involve five categories, and like, sorry about that.

But it's actually the fundamental idea of a statistical game being like a Bayesian lens, which describes like the the inference process, maybe like a parameterized Bayesian lens that allows you to learn.

But a Bayesian lens that like comes with a loss function. That's still the same idea.

And it turns out that loss functions like the relative entropy and the free energy, they actually have a really neat

categorical structure. So the things I was calling like free energy games before, or like

relative like Bayesian inference games, those are actually now a lot even neater still. So that's really nice.

And the way you get predictive coding is still as like a functor from some from these like statistical games, to dynamical systems. So like that idea is still the same. It's just actually it turned out that the statistical games

themselves, there was actually a lot more nice stuff going on there than I had realized before.

Awesome. All right.

You cite Erisman and Van Bermisch work on memory evolutive systems. To what extent have you used their work in the thesis? Would it be correct to say that there might exist a functor from that book to your dissertation? Or indeed, to any of your citations? Is there work someone could do to explore the connection between the two? To me, this is a question of one instance, if a learner or researcher performs active inference on the interface between a work and a citation more generally? Great question.

Yeah, I think it's a really nice question. And actually, I, so I read the this work on by Erisman and Rambamish on memory evolutive systems, like a few years ago, and I found it very like abstract and confusing. And I was like, yeah, this is this is all really nice. But like, how does it actually connect to the brain and how things like brains work? But I've, I've kind of softened my stance on it recently.

And I think actually, there's some really cool stuff in that work. Some really great ideas about like hierarchical complex systems and emergence, and like, even things like neural coding.

And it's kind of, I think, like, slightly in danger, I think, of being forgotten, because people are perceived and think, okay, well, like, how does it actually make contact with things I know about now? Things that like actually scientists use. And so I think it would be great to explore the connections, like, in a kind of more detailed way. I don't know precisely what that would look like. I heard a talk, actually, by Andre

Erisman. I think it, I don't know, I don't, there may be some on YouTube, I don't, I don't know. This was in consciousness, mass of consciousness, like workshop at the end of April.

And she explained some of the more recent stuff they've been doing. And it sounded like actually

they, there is a proof, I think, that heavy and learning does actually give you sort of what she calls cat neurons. And so before I've been like, okay, well, it's a nice story. But is it does it does it actually, it doesn't actually relate to things like heavy and learning? Or is it just like an analogy? But it sounds like it's actually a bit more precise than I had thought. And obviously, because of the connection between things like heavy and learning and predictive coding, it sounds like, you know, there, there should be a way to relate something like predictive coding to memory evolutive systems. And that would be really nice. Because I, you know, sometimes think about this kind of direction that I'm trying to go in. So, but try to think about, like nested active inference systems, like, you know, you might have a little region of the brain, and all the cells are doing, like, active inference, and the circuits do predictive coding, and they make up a brain, and the brains make up a, you know, society and or an organization. And that story of like hierarchical complex systems is really the story that this work is trying to tell as well, just in a different kind of way. And so I think there's some really, it's a really fruitful area. And I really hope, you know, somebody does dig into it. Because it will be because I think it's, I think it's great work. And I think it's just work that was kind of maybe ahead of its time. So it kind of got left, you know, in by the wayside, maybe a little bit. But I'm really glad to see somebody asked about it now.

Cool. Just one short remark on this. And then I'll hear Dean, please. It's almost like, when people draw associations, like a line of sight, Oh, well, emergence is like this, it's like, they're surveying, and can have quite far vision. But category theory is like building the high speed rail, where it's like locked in. And then we can say, there really is a relationship of this exact formal type between heavy and learning and predictive coding, let's just say some type.

Mm-hmm . So then that can be really relied on as a foundational building block. And it kind of brings those literatures and merges those streams, or at least adjoins the streams or naturally transforms the streams or whatever the appropriate terminology would be. But in a really solid way, that's not just like, well, one person on a mountaintop saw that these two things could be connected. It's more like an engineering project that really locks it in for everybody. Yeah, I think so. And I hope, you know, some people are incentivized to do that engineering. I think that'd be great.

All right. Last of the pre-submitted questions.

So in the Par Pizzullo and Friston 2022 Active Inference Textbook, the first question in the recipe for designing an active inference model, this is chapter six, is which system? This entails three pieces of data. One, the agent, generative model. Two, the interface, sensory data and actions. And three, the external environment, the generative process. So how do these relevant big chunks of notions from the recipe for making an active inference model in the part-of-the-part-all textbook map to the sections of what you've done in your dissertation?

That's a nice question. And it like takes me on to like a question of like action and active inference beyond the sort of approximate inference that I was talking about in my thesis predominantly. Maybe here I could like draw some stuff. And so I don't know how easy that is for you, but I can share my screen. Yep. Let's see how this will work.

Yeah. Okay. Share.

All right. Can you see my like cursor thingy? Yeah. Okay, cool. So, um,

so I like to draw this picture and this is like the little channel C, which does prediction.

And then, and it goes like from here and it goes like outwards. And this is the inversion, the like approximate inversion. And it takes in like a prior, which is comes from like here.

And it takes in like an observation. So we could think of this as like,

um, this is like high level

belief.

And like lower level.

And this is like,

somehow you get something which gives you an observation.

And then this gives you like,

new belief, high level.

And this thing is a Bayesian lens. So that's like what the Bayesian lenses are all about.

Um, and like the thing that I was talking about in my thesis is, um, doing,

you know, turning this into like a predictive coding dynamical system.

Um, because you've got like neurons, which do this sort of prediction part and then neurons,

let's do this like, like, um, passing back of error signals to do this updating part. And they have

this kind of like bidirectional form. And that's, that's really neat. And so like, you can like put in

another one here, another one here, or you can just start with your prior, which is like something that I often write as π up here. And then this gives you like a new π at this point.

And so, you know, this is going always from like high level beliefs to lower level beliefs. And down here, you might eventually get to like the lowest level, which is like, you know, if you're thinking about visual systems, then this to be like, you know, the V1 visual cortex stuff, or like the stuff you get in like the retina and you get like prediction about, let's say visual signals, but then the world, you know, so you get this predicted, but then the world comes back and it actually gives you like actual, this is predicted,

gives you actual signals, and then the brain uses that update its beliefs up here. And so really,

like the interface is somehow like this. Okay. And so this is, this is like lens, lens world.

And this is like external world.

So that's where the like external environment comes into the picture. And you could think of everything

in here as like agent. But this doesn't really tell us much about action. And it doesn't tell us much

about like interaction. And so there's another kind of story for that, which is kind of similar in many respects, which is the story that is told, at least in category theory, by what's known as

categorical systems theory. And there you get pictures like this. So this is like your like world. And

there you've got like a sound like system inside the world. And here you've got another system inside the world. And so people then draw like things that are called wiring diagrams to kind of represent how these things interact. So for, let's call this A, for A, maybe there's another system over here.

And this is B, and maybe that's C. For A, its environment is like everything that isn't A, it's, you know, B and C.

And so maybe like it's connected like this, I don't know, some like weird like wiring thing.

Actually, maybe that maybe that should go like around to here. And then this goes like to that.

I don't know, something like this. So this is this is just what's known as a wiring diagram. It just tells you how the systems are like coupled together. And that would be like specifying like how the information flows across the Markov blankets of your systems. And so like, this would be like the, this box is like the Markov blanket of A, and then this like wiring is like the data of the coupling of the blankets. And so there's a whole theory, a whole story in category theory about how to sort of compose, if you've got like, you know, if you, you know, it has a composed dynamical systems of this kind.

And so, you know, what we do is we like say, okay, well, we write down like the type of, if we say, if we call, if we write I A to represent like the interface, like the data of like, this is like X, and this is like Y or something like that, the types on the, the kind of shape of the Markov blanket.

But then what systems theory tells us is that for each like interface, we get a whole category of systems that you could put in that box to like animate it to make it like a living system.

And then, you know, this is like a closed box at the moment. And so maybe it just represents like the whole universe, which doesn't have any like external environment. But you might also have like, you know, something like a company. So you call it like company X. And that has like, you know, some agent which decides what the company does, or some agent here, which like get takes in information from the environment. And so this, this, this wiring is then something that gives you a morphism from like A and B and C to X. And what this like, dynamical, like systems theory tells you is that there's a, let's call that W for wiring, is that there's then you get this functor from $\text{DIN } A$ and $\text{DIN } B$ and $\text{DIN } C$ to $\text{DIN } X$, which tells you then how to wire all these dynamical systems together to give you one that fits into this outer box.

And so this is like tells you how to do all this for just dynamical systems. But of course, we're kind of interested in like active inference systems. And so here you don't just have like some like, you know, DX by DT thing to go in here. You've got, you know, the data of prediction and, you know, plans and goals, and maybe the maybe then something about how to like, you know, actually do the inference like the statistical game stuff. But really, the story is the same. And so if you have, you know, this, these boxes again, what we can do is if we have a box like this, X to Y , is we can think of the, like the, an agent as having as a part of it, a way of predicting, say X , we think of, let's, let's call this something else. I'm sorry, let's call the, let's call these S . So this is like the fence data that goes into the system. And then there's call this A . And then this is maybe like the actions that it takes. Let's, let's now call this box. I don't know, B.

Um, oh, no, I don't like B. Let's call it Y.

So now let's think, okay, well, part of what an active inference agent is doing is that they, you know, they're predicting, they've got some, some like high level beliefs of type Y. And then they make a prediction about not just the sort of sense data, but also about their actions. And then,

you know, their body like samples from this belief about how they should act, or you've got these little cells, these little like mechanisms, which try to correct the prediction errors to like make that those actions into reality. And then maybe that gets like coupled to some other system, which receives those actions as inputs. And, you know, you might have another part here, which, you know, goes in the other direction, which says, okay, given my actual sense data, and, you know, my actual actions, maybe my body is still learning how to like play tennis or something, like what, what, what are my new high level beliefs? And so you can do this whole like story again of Bayesian lenses. Um, and this goes like that, and that's, you know, and this is now just why Y, and then this is like S times A, and this is S times A. So you can tell this story, and you can build like hierarchical systems. But now you have like a way of filling in each box. And in principle, a way of coupling them together to give like, you know, composite active inference systems like companies or like, you know, colonies of cells or things like that. This isn't a story yet that I've written down. But I know it works. I'm in the process of slowly writing it down. So the kind of fundamental structures that are in the thesis are like part of the story. The thing that's the main thing that's missing is kind of like reorienting. It's just like reorienting this picture so that it's not like going from, you know, here from left to right. It's like sort of poking down into like the box. If you see what I mean. And this is like, this is like the stuff you put in. So this is like your lens thing here, and you sort of fill in the box with your lens. So I think that tells, I mean, that tells you how like the chunks of the thesis kind of fit into the standard active inference story. The main job, at least as far as I'm concerned now, is to like, spell out the details and then start to talk about things like what happens if like, you know, we've got Y and you've got Z here. Like what if, you know, they're trying to form part of an organization, but like they disagree about what to do. Like how, how does an organization resolve disagreements? Like what if you've got like two signals from your brain to your muscles that tell you to do different things? Like how do you resolve that kind of, that kind of like, you know, how do you get like harmonious active inference systems in some sense? So I will, I'll stop sharing my screen now and then we can sort of return to the discussion. But that's like, again, my kind of long-winded answer. Awesome.

I can share the pictures. I see, Dean, you said you'd shared something, something, but I can share some more things.

Dean unmute and then go for it. And then Ali.

No, I just shared something from about nine years ago that I was doing in my programming that everything you just wrote out was, I just wanted to share one part of it that confirms that you could, you could come from completely different places and arrive at the same destination. The whole bi-directionality, the whole paradigm, predictive parabolic. Yeah, this, you can bring this into classrooms and have it make sense. It doesn't, doesn't have to be esoteric or too, too remote, I guess. Yeah.

What I mean, that's great to hear. I'm really glad to hear that.

Yeah. It's really awesome.

Thanks. I mean, yeah, as I say, there's lots, lots left to do. But I think an important part of that is spreading the word. It's really great. But yeah, that process is beginning, thanks to you guys.

So yeah, that's good.

The rehearse, play, reflect cycle actually begins when you reverse the direction of thought.

Reading from Dean's email. Interesting.

Self-organize. Yeah. It's all part of this whole cat, but this is the thing.

Sometimes the, like, I love the questions that we've had so far, and I want to hear Ali's questions.

And then I want to be able to take it and have people not be intimidated by it, because you can really find yourself doing these things on a daily basis. You just don't necessarily identify them the way that we are. But that doesn't mean that we have somehow lost access to the things that we're discussing today. So that's, I'll just leave it at that for now. Yeah. Cool. All right. In the rest of the time that we have with you for the .2, Toby, Ali?

Okay. Thank you.

Your video's frozen. Your video's frozen. Maybe you can just turn it off.

Yeah, go for it. Thank you. All right.

Sorry.

Sorry.

I noticed in your thesis.

Sorry, Ali. There's just a lot of, a lot of, like, yeah.

Yeah.

What would you say is the chapter six equivalent? Is there yet a recipe for designing the kind of categorical, kind of compositional organization that you alluded to?

I can answer that quickly.

Yeah. Go for a quick answer. And then I think now it's better, Ali.

Yeah. So, okay.

The quick answer, well, it's a quick answer, but it's not a quick thing to do.

But there's a great book on categorical systems theory in its kind of great generality, which doesn't explicitly talk about active inference like I've been, but it does talk about this kind of composing dynamical systems in this hierarchical way idea. And that's the book that's currently being drafted by David Jazz Myers. I can circulate a link to it.

M-Y-E-R-S.

And yeah, so that sort of spells out the kind of general structure.

M-Y-E-R-S.

And I believe he and Matteo Capucci have got a paper upcoming at the upcoming ACT conference on applied

category theory this year, talking about using that framework to do kind of cybernetic like things.

M-Y-E-R-S.

And so what I've been thinking about lately is trying to fit active inference and the stuff I've been doing into that framework, which has been sort of developing in parallel.

M-Y-E-R-S.

And kind of fortunately and maybe slightly unsurprisingly, they fit very nicely together.

M-Y-E-R-S.

M-Y-E-R-S.

M-Y-E-R-S.

And so the pictures I was sketching are kind of a fragment of that other framework.

And as I say, I think they go quite well together.

M-Y-E-R-S.

Awesome.

Okay, I'll go for it.

M-Y-E-R-S.

Okay, sorry about that.

All right, so I noticed in your thesis that you don't rely too much on the notion of Markov blankets, but instead you use this notion of polynomial functors as its more formal or perhaps more rigorous replacement.

With regard to that, in your paper,

polynomial life, you say in the literature on active inference, sorry,

and the free energy principle, there's much debate about the concept and role of Markov blankets, which are sometimes conceived to represent the boundary of an adaptive system.

We believe that the algebra of polynomials will help to make such thinking precise, where it departs from the established usage of Markov blankets and Bayesian networks.

So would you care to elaborate a bit on that?

And how do you see polynomial functors as overcoming some of the limitations of the notion of Markov blanket?

Yeah, okay.

Okay, so I'll preface this by saying I think some people might disagree with me, but as far like I...

There is a notion of Markov blankets that for me is nicely defined, that exists in the sort of Bayesian statistics literature.

And it says if you've got a Bayesian network, you can consider the nodes in that network, and you can ask what are their Markov blankets.

And in particular, you can define the Markov blankets of a particular node to be its parents.

So those nodes that it's conditionally sort of dependent on.

Its children, sort of those nodes which are conditionally dependent on it.

And what are known as its co-parents, so the other parents of its children.

And that's a very nicely well-defined notion.

But it's only nicely and well-defined in this context of Bayesian networks.

And so it's like...

Generalizing that to kind of situations where you've got interacting dynamical systems is something that like informally makes total sense.

You want to say, okay, well, I have this thing in the world and it behaves so it changes in time.

And it's sort of a coherent unity in some sense.

And so it has a kind of boundary.

And because it has a boundary, like information and energy,

or it, you know, from the environment always have to flow through this boundary in order to interact with it.

And that sort of boundary notion, it seems, is like what is in other contexts also called a Markov blanket.

And I think if you set up your dynamics in a particular way,

you could, or your definition of Bayesian network in a particular way,

the two ideas that, you know, might coincide.

But I don't think that's true in general.

And I think what people mean in the literature when they're talking about Markov blankets,

but in this dynamical context, I think all they really mean is like boundary or interface.

So the data of the system's like capacity for interaction or something like that,

the stuff that is on the edge of the system through which interactions with its environment are mediated.

And so that's a general idea is something again that appears nicely in categorical systems theory,

but in a very general way.

And it doesn't require any of the notions of Bayesian statistics,

and it doesn't require, you know, conditional dependence or

like parents or children of a graph, because, you know, you might have something more complicated than a graph.

Like you might have like edges which are allowed to connect more than one node,

and that gives you a hypergraph. And there, of course, the notion of Bayesian, of Markov blanket in the standard sense,

just doesn't make sense at all.

But of course, the notion of interface does. And there's a certain like there are certain categories of interfaces.

And those wiring diagrams I was drawing before are an example of a category of interfaces.

Like they don't have any data of the systems themselves. The category just tells you how to sort of plug interfaces together.

And so I think it's kind of neater to think about this connectivity and the interfaces sort of separately from the stuff that fills the interfaces,

like the data of the dynamical systems or of the active inference like games that I've been talking about.

And so a really nice kind of category that you can use to talk about interfaces and indeed to talk about lots of other things,

but in particular to talk about interfaces is this category of polynomial functions.

And it kind of generalizes these wiring diagrams that I was drawing earlier.

And it does so in a particularly useful way, I think, which is that if you think of if you sort of formalize the

interface of a system using this polynomial language,

you can start to talk about interfaces that themselves might change depending on the context.

So like when you get into the car to drive somewhere, effectively, you've sort of changed your interface because now you've got this like power around you and like you use and it behaves in a different way than if you're just on your own.

Or if you're like just, you know, a normal, you know, normally going about your life, you might, you know, blink, right?

Or you might close your eyes. And that means that you're no longer receiving visual signals from the world.

And so like if you just say, you know, you're writing down your like active inference model and part of the model is that it receives visual signals.

Then, you know, talking about situations where those visual signals go away.

It kind of you can do it, right? But it's not very neat.

And so to sort of account for interfaces that might change like this, it's nicer to be able to say, OK, well, my interface actually is allowed to depend on what I'm doing right now.

So like, you know, I'm currently on a video call and it means I can do certain things, but like I'm currently wired into the computer, so I can't so easily go and walk around like the set of like signals I receive or actions I might take might change.

And so I think it's easy to start by saying, OK, when we're talking about Markov blankets in dynamical systems, we just mean like there is some set that is like the action set and some set which is the sort of sensory set.

And that's just for me, it's just not enough because of the fact that the world is dynamical, you know, context change, you know, coupling might be dependent on like you might be in an organization and there's some executive who.

Who's in charge of defining the team structure and that mean the team structure is like the wiring of the agents in that organization.

But it depends like that how they behave depends on the decisions that that executive makes.

And so like their Markov blankets are like conditional on those decisions.

And that's something you can talk about nicely in this polynomial language, because you're allowed to have systems which like affect the wiring of other systems.

But it's really not something that's easy to talk about in this like standard framework of of of like interfaces and boundaries that is dominant in the active inference literature.

And I think that's because really in the active inference literature, there isn't a framework for interfaces and wiring.

And that framework is supplied by these categories of interfaces.

And so you can just choose one and you can, you know, hope it does, you know, choose one that's appropriate for your situation that you're trying to model.

But it helps to actually have one that allows you to do this wiring.

And that's something that using category theory allows you to do because it's all about composition.

Awesome.

The textbooks will be rewritten.

One quick thing, Toby.

One quick thing, Toby.

Does the timing of the choice in your experience really matter?

The timing in what respect?

Whenever you do decide to make a choice, doesn't the effect of the timing, the whenness of activating that choice play a huge part in what follows?

In terms of the category?

Yeah.

Yeah.

So do you mean like if you're choosing to use a particular kind of interface language, say, or describe a thing using a particular with a particular interface or like particular kind of boundary?

Yeah.

so i i agree and that's why i kind of tend to work quite abstractly so rather than saying i'm working with a particular kind of interface language i'm just saying i will assume my interface language allows me to do these certain things and so that's not so useful for like actually implementing a particular model but if you're trying to think about like systems in general it's helpful to work like agnostically in that way and so the reason i sort of advocate like this for polynomial functors is i think that they are they sort of sit at a nice sort of intermediate level of abstraction it's not it's not to say okay well there is a collection of interfaces and some collection of wiring them together because that doesn't tell you like actually how that works um so it does it is slightly opinionated it's different from certain choices that people and other bits of applied category theory make um but it does generalize this simple case where you have just a single set of outputs and a single set of inputs or single set of actions and a single set of um like observations and so i think because of that it's it's a kind of nice it's a nice sort of intermediate level i think it's it's one i've like tried to work with because i think it's quite you know natural um but yeah you can sort of totally work at a very abstract level and that's what um david jazz does with his systems theory he just he's just like talks about interfaces in general like they don't have they might not have you might not have like actions at all um you know it's completely agnostic

uh okay thank you so much so uh my second question uh is about the relation between uh your compositional

account of active inference and uh the recent line of research uh done in bayesian mechanics namely uh the discovery of the duality between fep and uh constraint maximum entropy principle so

uh as discovered uh primarily uh through the work of uh dalton saktivadevel uh in the last couple of years uh there's this uh nice duality between uh the observer dependent uh the uh i mean uh the viewpoint from the inside observer of the agent and the outside observer so based uh respectively uh the perspective of fep as opposed to a constrained maximum entropy uh principle but uh dalton uh has demonstrated that actually those two perspectives uh can be unified uh through showing the equivalency between the bayesian optimality and james optimality so uh bayesian optimality from bayesian optimality we can obtain fvp and from james optimality we can um uh we can acquire um a cmap uh so given that uh the notion of bayesian lens also uh comprise both uh bayesian bayesian inference and uh its inverse process

uh uh do you think uh uh the bayesian lens as providing uh a kind of uh i don't know a tool to account for showing this uh kind of duality between uh fp and uh constrained maximum entropy principle i think the answer is probably yes but i think this is a really great question because i genuinely i don't know the story here like i mean i i know um a bit about dalton's work and i think it's wonderful um but i really don't know yet how it connects to the stuff i've been doing and it's it's really like an open question for me um but there are some like tantalizing signs that as i say i think the answer to your question is yes there is that kind of unification of the two kinds of story um and some of the ingredients of that story are kind of floating around out there so i know that there seems to be a connection or like i know there is a connection between quantities like um the relative entropy and like uh sort of infinite like energetic concepts you get in physics and through that you can make a link to the kinds of things that dalton has been doing with gauge theory and i i i feel like there there is a kind of geometrical structure that you can put on top of these statistical games that i've been working with because remember they are these lenses with these loss functions and these loss functions seem to have this kind of um as i say this kind of geometric thing going on with them but really like re like explaining or like putting matt's constrained maximum entropy into statistical games language um is not something that i've done and it's something that you know you know um i've been thinking about doing and you know i'd be delighted to talk to people about more um and so i don't know precisely the details um there is a kind of other duality which i think is quite nice which is related to this um and i think and i think again at the nexus of these is this like systems theoretic idea um because of course all of this work of dalton's lives in this kind of like dynamical systems world where you have like dynamical systems like that are interacting and information flowing between them um and there are some really tantalizing um connections again of course between physics and information processing and i i learned the other day for instance that there are certain classes of um generative adversarial networks um the sorry this is just a slight tangent but i think you might be interested to know certain classes of generative adversarial networks that you can you can set them up and the two like um the two like parts of the network tend to like instead of settling into like an equilibrium they seem to like exhibit some like orbits around each other so they don't like achieve some like equilibrium state in the game they're playing they kind of just like trot around um but somebody was telling me about um like if you have an orbit you could sort of in your efficiency you think okay well like what what is that orbit like like conserving is there like a quantity that it's conserving and in this case and i don't know the details but in this case um this person found that the quantity that was conserved in this orbit was the relative entropy and so like there's this weird thing going on these things are playing a game but they're still like at this kind of like somehow like steady state of relative entropy um and so there's this seems to be this deep connection between these kinds of like game-like systems and like sort of information physics in some sense um and i'm aware i need to head off at a moment but i

just want to draw one more picture and i'm sorry you probably had loads of questions but we kind of ran out
i haven't really managed to answer many of them but hopefully we'll get to the rest of them soon
um but if i just draw as i say one more um picture
which is a kind of slightly uh different kind of duality but a similar kind of similar kind of flavor
so what we have in systems theory is this category you have this category of interfaces and
that of systems and the morphisms tell you how to like wire these systems together to give like
hierarchical or like composite systems and you can if you have if you have something like this thing
which i wrote before in which i said you know for each interface i was like a category then you can
sort of think of this as a functor from int to like cat the category of categories and one thing you can
do with index categories of this kind is you can sort of turn them into what are called like
vibrations using what's known as the growth and deconstruction if you know about physics you've
probably heard of a concept called a fiber bundle which is just like a collection of spaces indexed by
points of another space and so what you can do is you can write this vibration which says okay well
for each interface i've got a category of inter of systems on that interface and they're all like glued
together correspondingly with the the wirings of your interface category and so you write this as like
another functor which kind of forgets about the dynamical systems and just returns the interface again
and so you have this vibration thing here and this is like where kind of like physics lives
um and so what i've been trying to tell this story about you know in this story about active
inference is that you have a similar one where you have um for each type of interface the category of
agents that might be that might have that interface like you know we're all humans and so we're all like
the type that might fit in like a human shaped box um and so we're all in that like particular category
and you know we come together to form organizations and that's another object of this category of
interfaces and another kind of agent category on those interfaces and so that gives you another one of these
vibrations and the thing that something like predictive coding tells you is it says okay well given a
certain story about a type of agent can i turn that into a dynamical system which like actually animates
it and so that's like a functor like okay so like this so maybe maybe what we could do is we could think
of this as like maybe like some version of biology or we could think of this as like you know we could think of
this as like body and we could sort of think of this as like mind and so like this is telling us how to
go from like a description of like a mental system to like a sort of embodied system and i think some
of what the free energy principle is trying to say is that you know for certain kinds of system there's a
way to turn your description of a dynamical system into something that looks like an agent and so the thing
i'm wondering is that whether there's like some like adjunction or maybe maybe this is like i don't
know i don't know which where we might write labels on these but maybe there's like some adjunction here
um that allows us to like compare these two processes of like you know turning a sort of mental system into
a physical one and then going in the other direction from a physical system into a mental one
um and so maybe this is also like where like you know information comes in
um and it's a kind of similar story in in category theory um where there's a sort of duality between
geometry and logic um that you that's sort of sometimes written like this you have like truth values

down here and you have like logic over here and you have like space over here and like there's a way to sort of like draw a picture like this that says okay well logic is somehow like dual to space in some like sort of mumbo-jumbo way i mean again i've run out of time but i think there's a story that people try to tell about like science in this kind of abstract world and i think it's a similar story to the name of something that's being told over here um again lots of details and i don't know precisely the connection with dawson's work but i think it's kind of pointing in this kind of direction thank you toby after you have revised and proceeded um let's have 54.3 and talk more about action active inference so thank you yeah great thank you so much for having me and i look forward to speaking again and answering your questions ali thank you so much that was great okay great bye guys bye bye well

wow i'd like to ask dean a question oh ali please make a remark and then i have a question sorry yeah uh you know uh i really think of toby's work as uh really provocative as it uh as it really opens up pandora's box with many many uh different fertile questions so uh i'm i'm a firm believer uh this old saying that uh a sign of a great book is not how many answers it provides but uh actually how many uh questions it gives rise to so uh i really wanted to congratulate toby and said and writing such a uh uh i mean uh it it looks as if uh he he he began uh from the perspective of trying to address uh some of the uh i don't know some long-held questions but uh then it turned out to be uh much more deeper than that in terms of uh actually giving rise to many uh fertile questions so uh yeah that's really great uh

dean i i'd like to ask what is the difference or how can we think about being on time in time and out of time because toby said he was out of time with the answer and i i know that we've spoken about this when we consider some of the time consciousness and category theory work previously so what is it to be out of time yeah that's a that's a great question all i can say at this point is that it's easier to discretize when you're in time and out of time as a as a comparator than it is to say when you're on time versus those first two positions that's all i can say about that for now because i'm sure toby from a from a diagrammatic perspective could probably tell us what is happening when he's actually using his stylus versus not

right like the result of that leaves something in and out but while he's actually on on point he he he's literally in a different place so here's a here's a here's an interesting thing back to the provocative ali in the dot one was talking about virtual and actual models and was probably saying it in a in a way that's completely different than how i use those those same i believe those same two models are in effect and so um

um if you if you for example um draw up a play say in sports because we're we're talking a lot of strategy while toby was on today and a lot of gaming gaming theory and stuff you draw up a play and then you confirm that you have all the elements or resources that you need in order to be able to carry that out and then you're you're as a group communicating with one another like we are doing on this on this live stream right like you're literally calling out to play in in north american

football you'd huddle up and literally call out a sequence of code and then you'd have to literally reverse that and turn the inference as as a the shortest time window into a prediction like you would literally line up and then you would carry out this the simulation in actual terms this is all of that together means that i believe that the virtual version of the model is as toby was saying in the bi-directional sense going in one direction and that actual model is going in 180 degrees the opposite direction and then when he mentioned the dagger symbol today i was just like okay where does that fit in to this bi-directionality and the difference between the the sim the digital the virtual and the actual so these are the things that literally pop up at any at any given moment aren't necessarily part of what you consider to be the space that is inhabited by virtuality and actuality and yet there it is it just appears and then you have to sit down and figure out well where does that categorically fit so again don't want to go off on a on a deep rabbit hole philosophical take on this but i just find that at any given moment there is again so much math in this philosophy that it's it's kind of staggering and just to be able to have somebody who can explain that and draw it out is pretty impressive so i don't know if it's provocative what i just said i hope it isn't because man i did this work for almost a decade and man i think it's a very important thing to do is to be able to do this work for almost a decade and it seems like young people weren't scared by it maybe older folks are but i don't know uh so if i may add a brief uh comment on just what you said uh i was recently watching this uh talk by uh john bayes about the history of applied category theory uh so uh he had this nice tag diagram showing uh the uh the relationship between the different areas of knowledge uh i mean going from business to pure mathematics and uh he he claimed that most people in category theory or in most other areas uh consider category theory as being on uh the top of uh top of the pyramid as being the purest of the purest so uh as being even a kind of uh being on a meta level than uh even the pure mathematics but then uh he said that i believe category theory can actually be seen as kind of permeating all of these levels so it's not something that sits on top of uh the uh the hierarchy it's something that actually can be seen as influencing and permeating all the levels so that at least uh from the from the perspective of applied category theory uh it seems to be uh if not self-evident but something that uh gathers much i mean uh more evidence as the research goes by so i'm really optimistic in the in the research direction that applied category theory is going and hopefully in near future we can see it uh applied to many many different areas many many different unforeseen areas so yeah i think i'll just tag on to that ali if you if you use an abductive logic lens then it's simply a question of when did you enter on what point did you enter the abstracted timeline because if you see it that way it's not just top down and it's not just bottom up it's back to my question well to toby was does does the whenness of this really really matter if you want to be able to take it off the shelf and use it and that of course his answer was yeah if you if you want to deal with idiosyncrasies you're left with no choice but actually to acknowledge that the the the point in the continuum where you take this up but i think that actually makes it quite easy to see it permeating every level but again it's the it's the abduction the abductive piece of this back to last year in in august right we're coming up on one year since we talked about

how important that is and yet here we are um what ten ten and a half months later and it and it we're circling back we have another predictive parabolic on our heads you few comments on that it's reminding me of professor mike levin's work on basal cognition and the idea that intelligence isn't on the top of the pyramid either rather it's something that's distributed compositionally across levels and something that exists at all levels and so we can then still talk about higher order properties of systems or or properties of composites or aggregate systems while also grounding in the in the foundations um so yeah and that grounding and anchoring and stabilizing that's by itself is not a is nothing but a feature i think you also want to be able to though because there's a lot of talk today about being able to see this in in the dynamic in the way that things dynamically play out so again we're back to the what's the minimum two ways that we want to incorporate this observation both the stabilized and as searching right if we can if we can have one eye on each at all times it does take a bit of rethinking um there is a there is a there is an interoceptive piece to this what we feel from that is different so that brings us to the sensor fusion aspects of the uh dissertation yeah

7.1.1 bayesian sensor fusion yeah

a situation that is common natural embodied systems but which is not yet well treated by current statistical machine learning methods also i'll lead thanks so much for asking the markov blanket question i think his response is worthy honestly to review and unpack in in the um time we have we can just sort of review a little bit and think about what other questions we want to prepare for for dot two how would you ali restate the um limitations of the markov blanket concept for all the positive things that we've said in the previous 300 live streams and and where category theory can help address and and move past some of that uh yeah you see uh so this notion obviously is one of the most um uh or i don't know probably one of the most controversial uh concepts in whole uh fp literature because uh it has received a fair amount of criticism from uh many various perspectives uh but some criticisms uh uh uh at least as far as i know uh is uh they're they're basically directed toward uh the universality of markov blanket how universal uh is this notion as uh as it's claimed to be ranging from i don't know inert rocks to the brains right uh so uh does it apply equally for modeling uh inert rocks as well as uh uh come the most complex system we know of in the in the whole universe namely the brain uh so the response uh this line of criticism uh got from uh the researchers in the active inference area uh again to the best of my understanding uh is to the uh to the effect that uh although theoretically it can be applied to many uh i mean different systems uh with many different levels of complexity but then again uh for simple linear systems such as i don't know many uh non non-complex uh systems uh it it doesn't have anything interesting to say so uh rather than being applicable to all of these systems the relevant question should be uh it i mean there are enough insights we can gain from uh applying this notion to uh many different um situations and um systems so i believe uh toby's um alternative for this problem by uh proposing uh the polynomial functor uh as a replacement for markup blanket uh can cover this line of criticism pretty nicely it can address this criticism pretty nicely uh due to its

uh abstractness and uh a kind of more rigorous way of talking about both uh the universality and the individuality of each system so uh i think that's a a nice way to uh at least uh a viable uh i mean answer to this uh question that needs a lot more consideration so that's my take on it that's part of it okay and i'll i'll i'll give my um take two which is the markov blanket concept since the work of pearl in the 1990s is actually not doing that much philosophical work in and of itself it just describes the parents the children and the co-parents of any given node within a graph so first we're talking about maps not territories we're not talking about features of the world carving nature of the joints necessarily hashtag malandrews we're talking about the bayesian graph and secondly even on the graph it's not that a node is like by its essence an internal state or a blanket state or an action state if you use the first and blanket rather it's always in relationship to others so the concept by itself does very little work however people have deployed the concept implicitly and explicitly to describe interfaces of cybernetic entities the kinds of entities that it makes sense to describe as an active inference agent so we know from the uh particular physics and the strange kinds work the path integrals paper that we can describe using the formalism the rocks the simple active systems and the sophisticated active systems in order to have an interface concept that does justice to on one hand the relatively inert interfaces on the other hand potentially extremely conditional or contextual interfaces just having some uh tagging system of nodes on a base graph is going to get extremely messy it's not that it couldn't be done but just in terms of actually doing the programming it's going to be extremely messy you're going to need to have a lot of accessory or auxiliary variables that indicate what the current interface is at this time or what it would be if this happened and all these kinds of conditionalities would be like a less elegant way of describing what our category theory colleagues have been developing explicitly with the wiring diagrams and interfaces so just to kind of conclude there people draw the bayesian graphs and the particular physics is using a bayesian graph however bayesian graphs don't naturally have a lot of these properties that were really um being conflated with interfaces more generally and by using an interface concept as toby said at an intermediate level of abstraction that's slightly opinionated but also really flexible that is kind of the right coarse graining the right optimal grasp on the interfaces that we actually want to talk about for active inference agents and again just trying to get down to the firmware with the bayesian graph we're going to get lost in the details and any accounting system that we develop is going to be ad hoc and certainly less principled or powerful than what the category theorists start developing yes to ask you a question and i'm really curious what you what you think of this if if we're talking about the relationship in in with the interface could we could we see and i'm really asking this question i don't know i'm spitballing could we see the markov blanket as the more stabilized version or form of the relationship and see the polynomial version of the of that is the search version of that relationship what do you guys think of that i'm really and i'm really asking guess the first just listening to both of you talk here i'm just

wondering if i'm having one of those unitary moments myself or if i'm way out to out to

i'll give a first first thought then i'll also give a speculative answer

uh yes so uh one thing that comes to mind is you see sometimes uh people consider markup blanket as being a kind of uh static boundary or interface uh that just separates uh the agent from its environment but uh actually what it really is is a kind of uh providing a really dynamic

interface between uh those two internal and external states so uh it's not a static or stabilized in any way but of course uh in some situations we treat it uh as if it's a stable or i don't know static uh so

i mean coding maxwell ramsted uh every theory of markup blanket is inherently processual so there isn't any

i mean static uh uh i mean static uh conception of markup blanket uh at all uh but on the other hand

well obviously for the polynomial functor i mean not even not just polynomial functor but the whole

uh area category theory deals with objects not in their static sense but rather in their um in the

meaning they acquire through their relations through their morphisms and it's not about uh so one of the

main differences between category theory and the set theory is exactly uh i mean uh hinges on uh that

distinction between the static viewpoint of mathematical or i don't know even non-mathematical

objects and uh the the inherently uh dynamical or relational identity of the objects as defined in categories

uh so i believe polynomial functor functors uh again uh by themselves are um not are i mean inherently and

intrinsically dynamical processual and uh in in the terms of some continental philosophers such as alan badu

uh there they acquire existence through their imminent relations they don't uh they they don't need any uh

i mean notion of pre-existence as uh some of the platonic notions of mathematical objects and ideas

uh so yes in this sense i think uh polynomial uh functors uh i'm not sure which one can be seen as i don't

know more dynamic or processual than the other but uh at least to a degree both of those concepts can be seen

as uh sufficiently uh processual in my view awesome okay i'll just add with a few other um associative

comments one is to blanket something is to dampen and stabilize it whereas polynomial seems to invoke

like many names polynomos polytopos so potentially there's a multiplicity in the polynomial

that is not acknowledged uh so lexically in the markov blanket and then a more formal way to say that

would be a markov blanket might be like a specific realization of some little neighborhood within the

polynomial functor so polynomial functor is a generalization of the markov blanket and so in that

sense it requires more searching and it is more abstract and it is more general but when you do

stabilize it down then the markov blanket formalism may absolutely apply but it will but wherever the

markov blanket formalism does apply like let's just say that we stabilize down to a computer system

and we have an api and we consider that that's one of our blanket states that would also be seen as an

interface of course which is how people would talk about it and so it would be a polynomial functor

and so this helps us look back at the literature and see as has been done many many times in fep and other

literatures recent developments enable previous developments not to be invalidated but understood

as special cases or or constrained or stabilized cases and it only makes sense that we would start um

well putting aside what makes sense in active inference it seems that we are taking both paths

in some cases leading with the stabilized and generalizing but in other situations making unstable

generalizations

that then um precipitate stability inside of them

so more bi-directionality

great well we will have a 54.3 with the focus more on action are there any other topics that you

fellows would like to raise or if anyone in the chat wants to ask a question please go for it

uh no just i wanted to ask toby about uh the relation um the relationship between uh his uh theory or

compositional account of uh statistical uh inference uh as opposed to uh the uh the category theoretical

states um um states um states um states um states um states um states um states um states um

in one of the papers from 2011, I guess.

So, yeah, that's one question I wanted to ask Toby.

And, of course, a subsequent question about the relation of this distinction slash non-distinction to the accounts of quantum FEP.

Because in quantum FEP, Markov blanket is kind of replaced or rather blended with the notion of, I mean, holographic inner screen.

So how can we see quantum FEP as arising from quantum Bayesianism rather than from, I don't know, predictive quantum theory and the relation between those two frameworks of Bayesian inference to that topic?

Awesome. Yes. The quantum category theory, active FEP triangle.

And also, I think it's the question about reading the literature in terms of the interfaces between a work and the citations, I think, is extremely rich.

It's a really novel way to perform bibliographic meta-analysis, but also just in our own learning journeys.

Following this category theory, victim of knowing a thing by its relations, understanding the paper within the web of the past papers, which it cites, all of which are past and actual.

And then it's future time cone of possible relationships to things that haven't happened yet.

The co-parents haven't even happened yet necessarily, but that becomes part of the evaluation of the significance in the moment.

And if we could have the right generative model and understand how much attention to weight onto different things in the moment, that would be like being able to see the whole time cone at the right core screening.

But of course, as limited nest mates, we can't do that.

But we have our rules, our heuristics, and our approaches that address that nonetheless.

Anyone in the chat or Dean, what would you like to explore in 54.3?

I am just going to follow the two of you.

I got two big workhorses here.

I just got to get out of the way so I don't get trampled.

Let's look at the table of contents.

Here we're just shifting to the code of document, the back end that we use to coordinate a lot of our work.

Just so that it can all be seen in the video.

Chapter one is short.

And then Chapter 7 closes it out, but really opens it up with the compositional cognitive cartography, sailing the three seas, society of systems, more alliteration,

and many, many, many connections to areas of interest for all system scientists and complexity enthusiasts.

So, what a tour de force and a journey.

And it's going to be awesome that we'll get to see some of the next increments in the work.

Talk more with Toby.

Yes, Dean.

Can I just throw this in?

And it's going to indicate how old I am, but a long time ago, the first time a needle dropped on the first Van Halen album,

something changed about everything that we thought about guitar playing.

And if you weren't there in that moment, you wouldn't know how everything changed.

I'm not sure what category theory taken up on a mass scale will do,

if it will be the equivalent moment for all people who use math in all the different ways that they use it.

But I think that we are at the beginning of something.

That is, it is going to change things.

And I don't know exactly how it's going to change things,

but there will always, I think this will always now be included as how we try to both figure things out and out the figures in that three-dimensional space.

And one of the things that I thought was interesting in the dot one was when Toby said, well, I work in 2D space.

And now I'm curious about, I wonder what that would look like in a 3D space.

Well, we exist in a 3D space.

And so I think the applicability of it is going off at any given moment.

So, but I do really think this is the start of something.

I'm not sure exactly what, but I don't think it's, I don't think we'll be able to unlearn this.

I think it's going to remain influential for a long time.

So.

For those about to categorify, we salute you.

Yeah, right.

There we go.

Thank you.

You can always make it better.

Any other thoughts on this?

Otherwise.

Great.

Yeah.

Ali, please.

With any last thoughts.

No, I just wanted to say I'm really looking forward for the next installment.

I hope it happens soon because, again, I have so many burning questions.

I try to, I mean, whittle it down to some, I don't know, not too much time consuming questions.

But, yeah, I'm really interested in, I'm curious to know Toby's answers to them.

So, thank you so much for everything, for organizing this and inviting Toby.

Oh, absolutely.

All right.

Thank you, everybody.

See you in 54.3.

Bye.

Thank you.